



SQL DATA WAREHOUSE

Project

SQL Data Warehouse Project





SQL Projects



~Organize, Structure, Prepare~

- ETL/ELT Processing
- Data Architecture
- Data Integration
- Data Cleansing
- Data Load
- Data Modeling



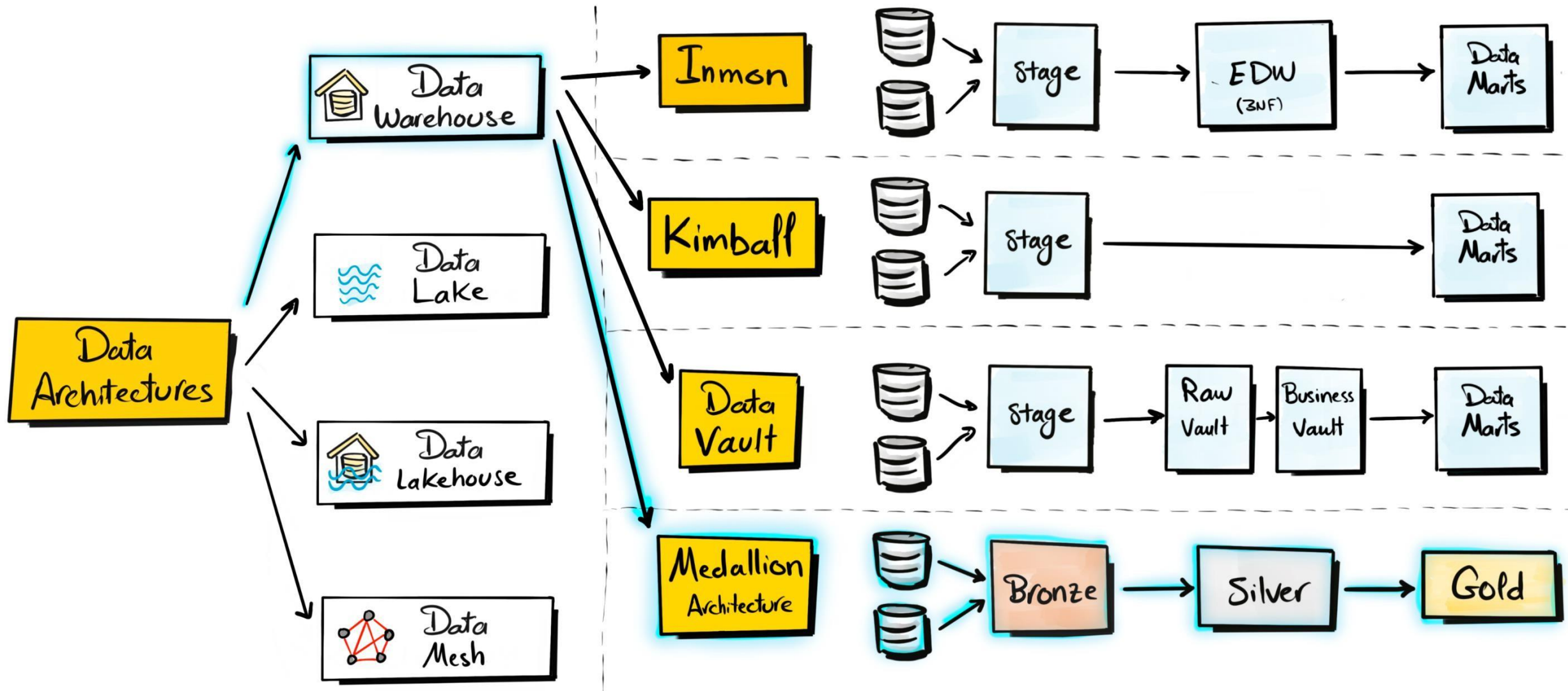
~Understand Data~

- Basic Queries
- Data Profiling
- Simple Aggregations
- Subquery



~Answer Business Questions~

- Complex Queries
- Window Functions
- CTE
- Subqueries
- Reports





Bronze Layer

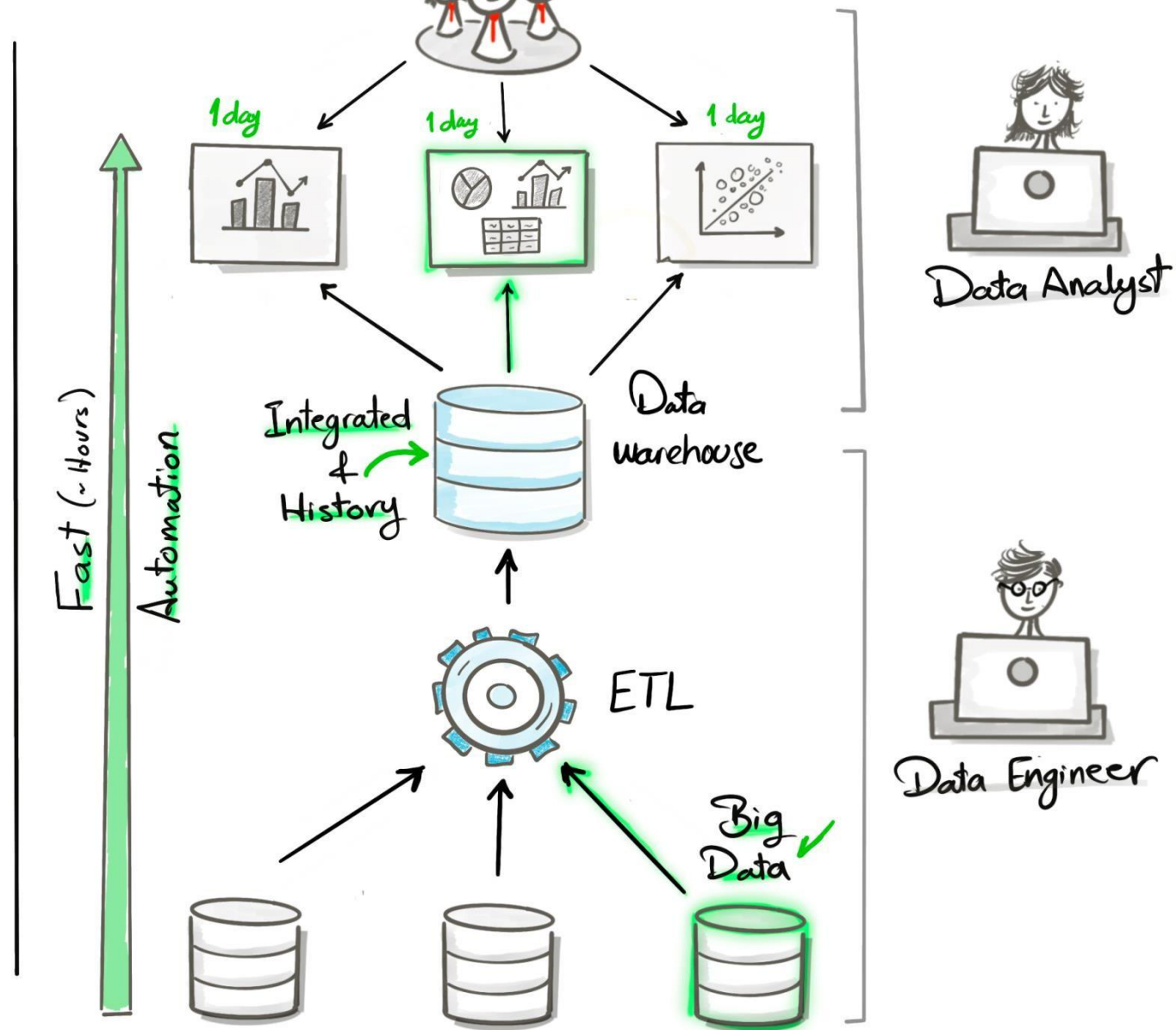
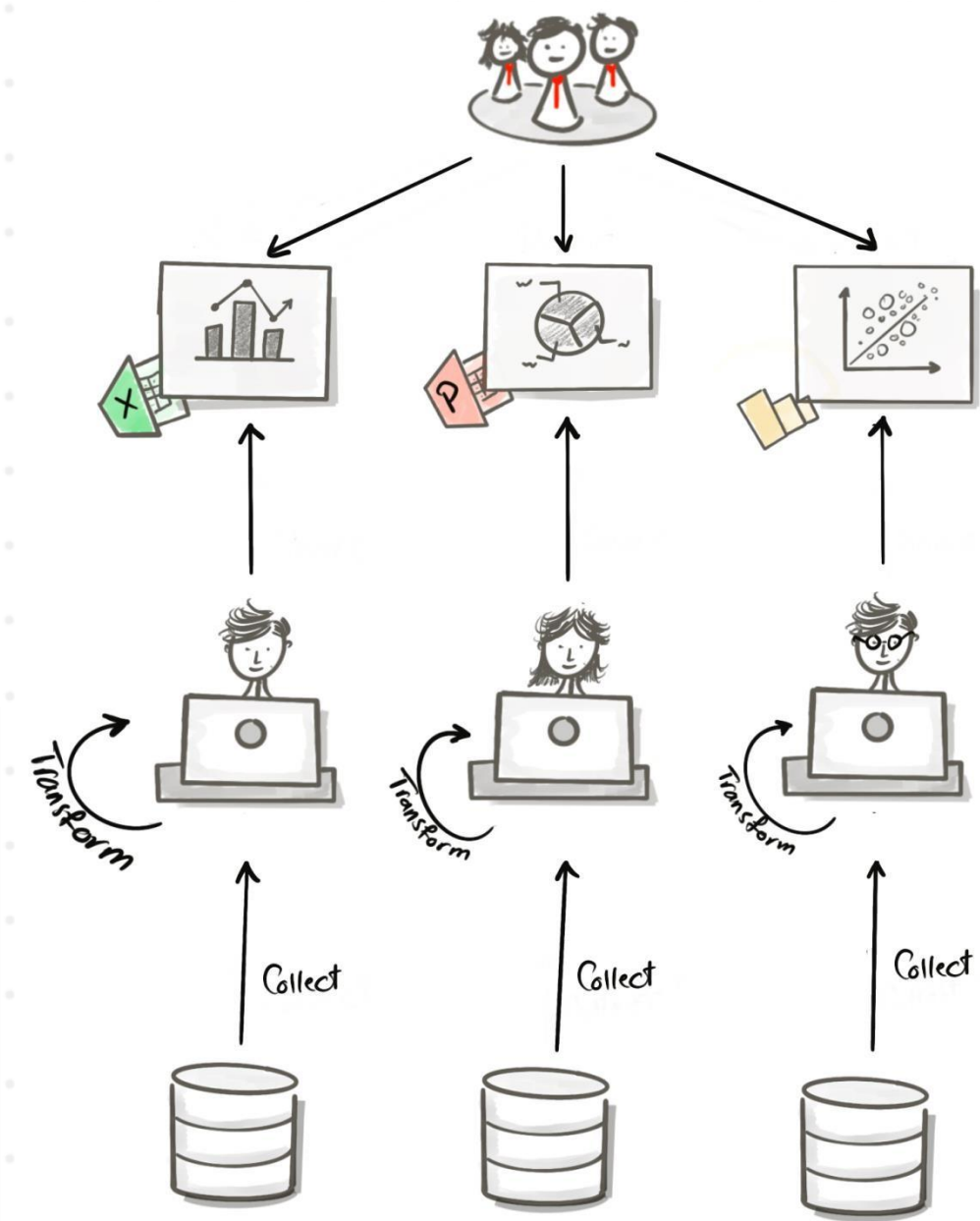


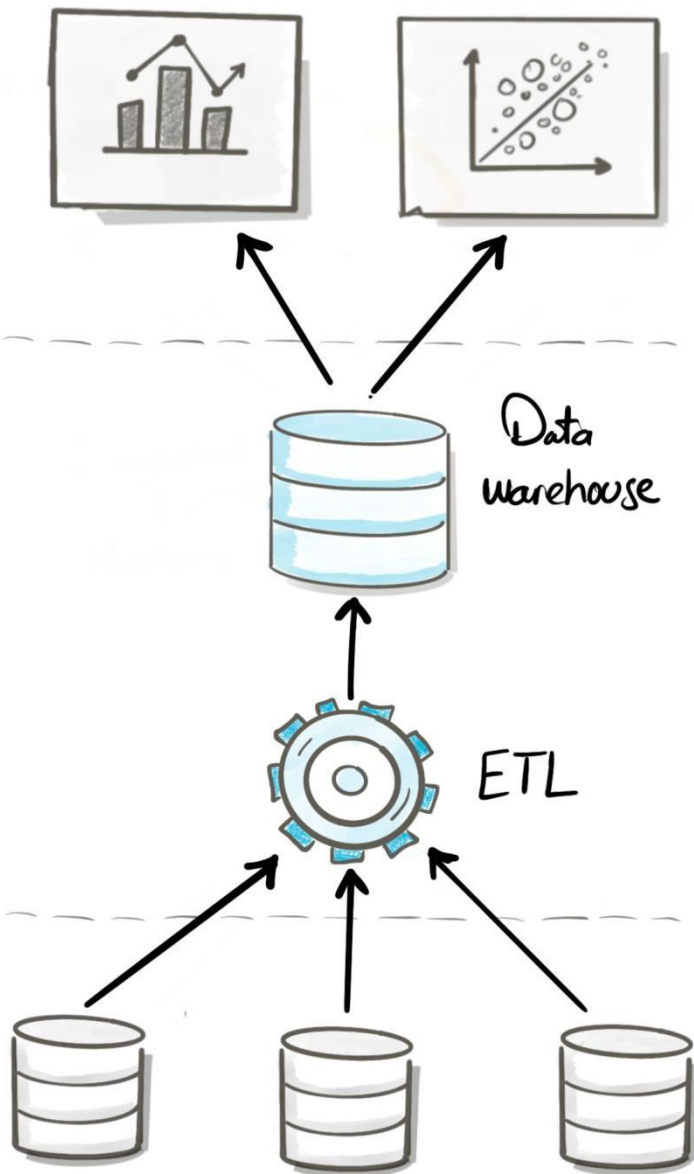
Silver Layer



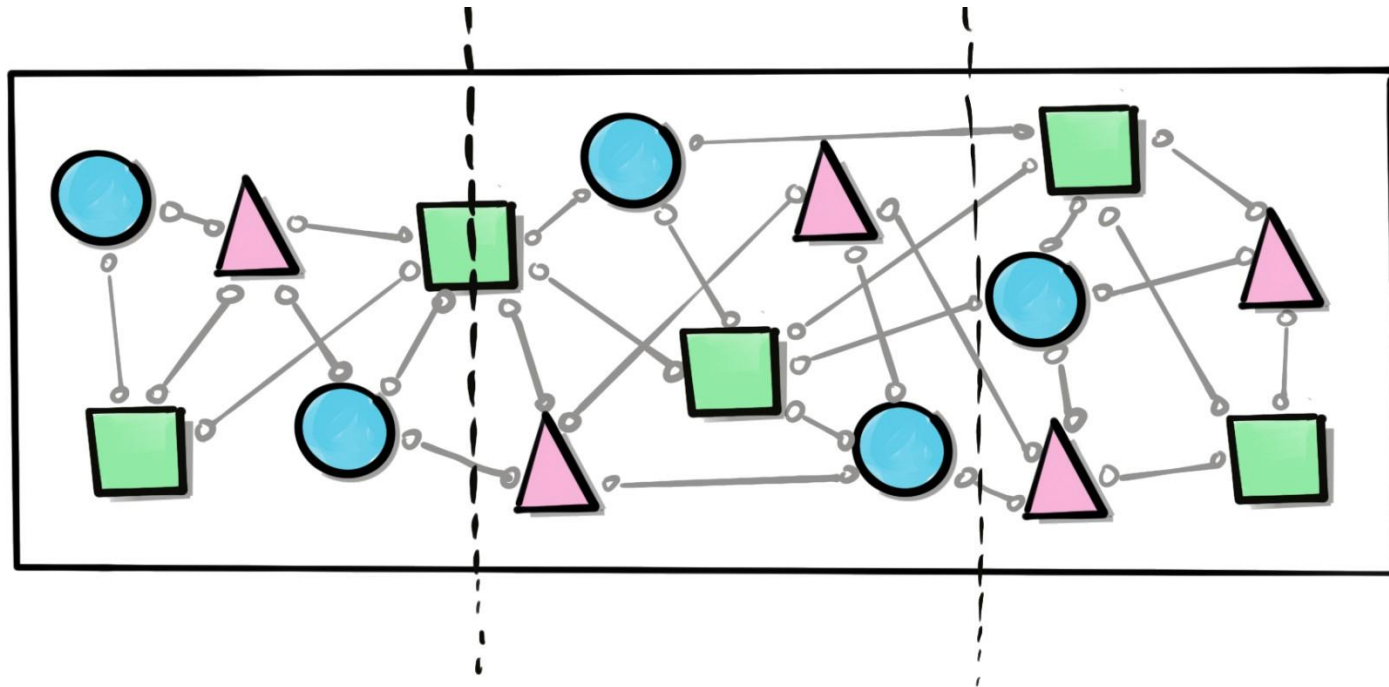
Gold Layer

Definition	Raw, unprocessed data as-is from sources	Clean & standardized data	Business-Ready data
Objective	Traceability & Debugging	(Intermediate Layer) Prepare Data for Analysis	Provide data to be consumed for reporting & Analytics
Object Type	Tables	Tables	Views
Load Method	Full Load (Truncate & Insert)	Full Load (Truncate & Insert)	None
Data Transformation	None (as-is)	- Data Cleaning - Data Standardization - Data Normalization - Derived Columns - Data Enrichment	- Data Integration - Data Aggregation - Business Logic & Rules
Data Modeling	None (as-is)	None (as-is)	- Star Schema - Aggregated Objects - Flat Tables
Target Audience	- Data Engineers	- Data Analysts - Data Engineers	- Data Analysts - Business Users

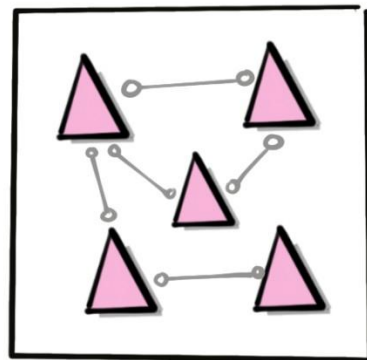




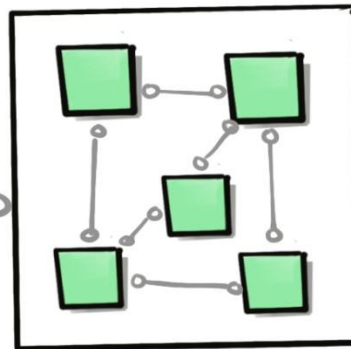
without
SOC



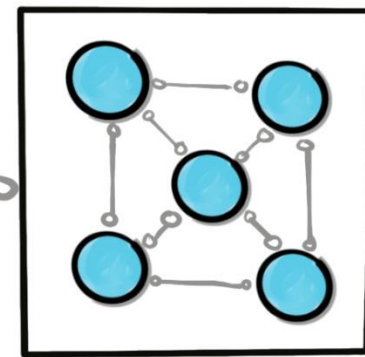
with
SOC



Module A

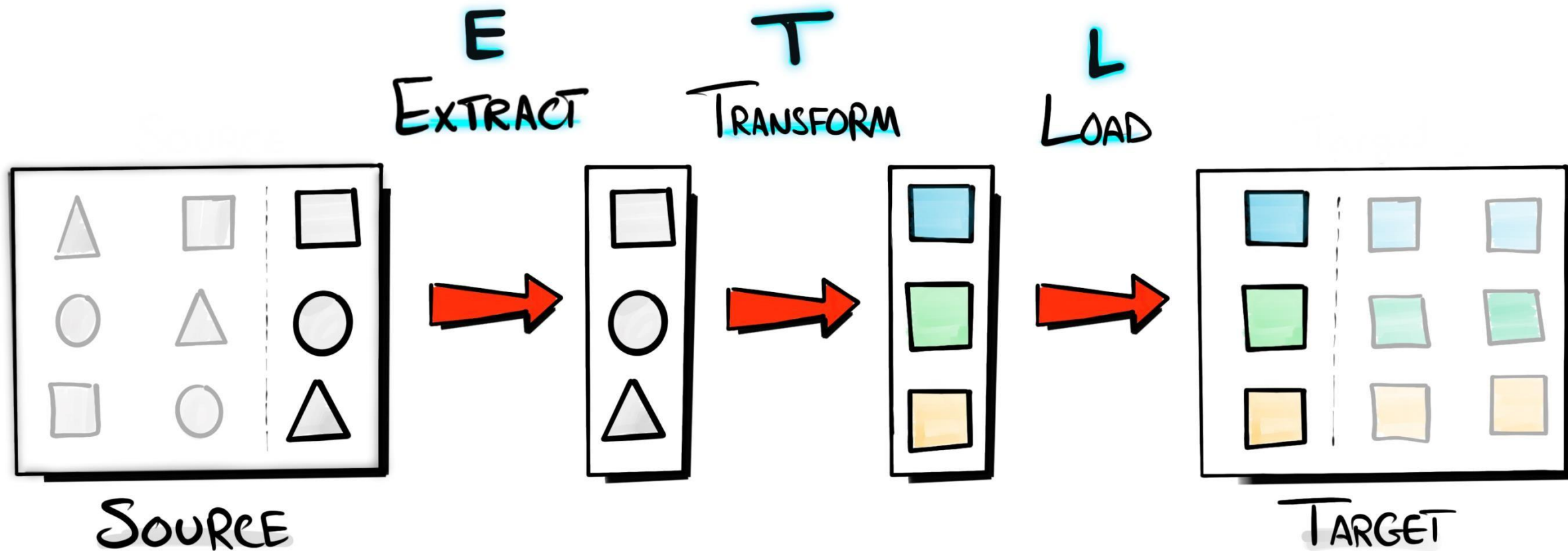


Module B

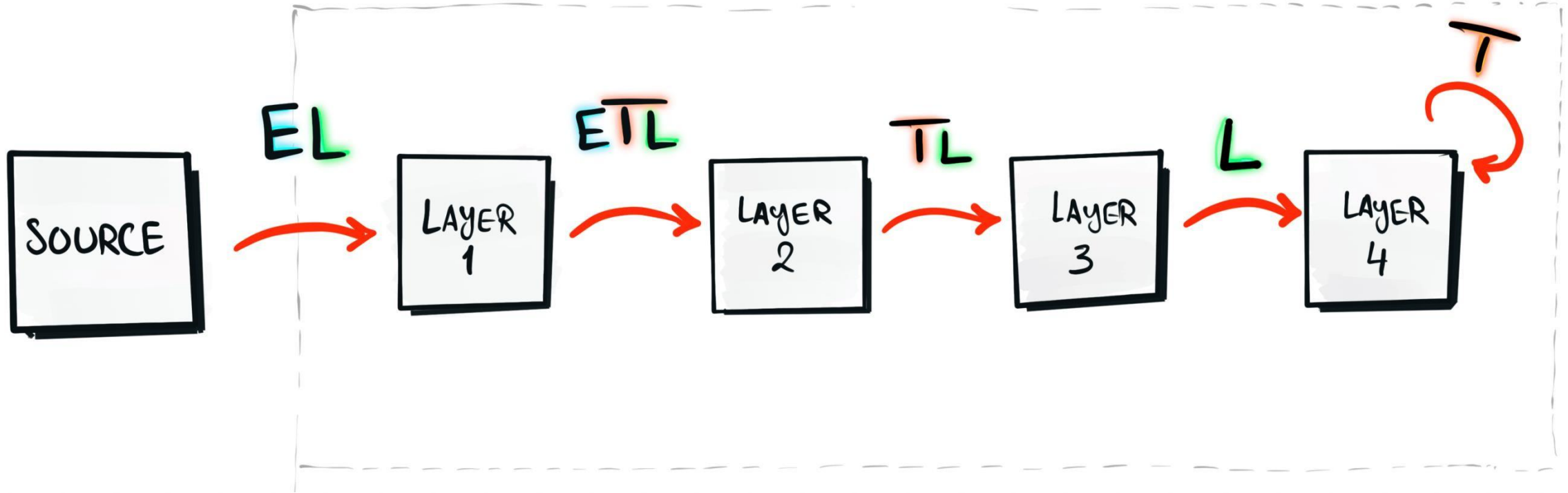


Module C

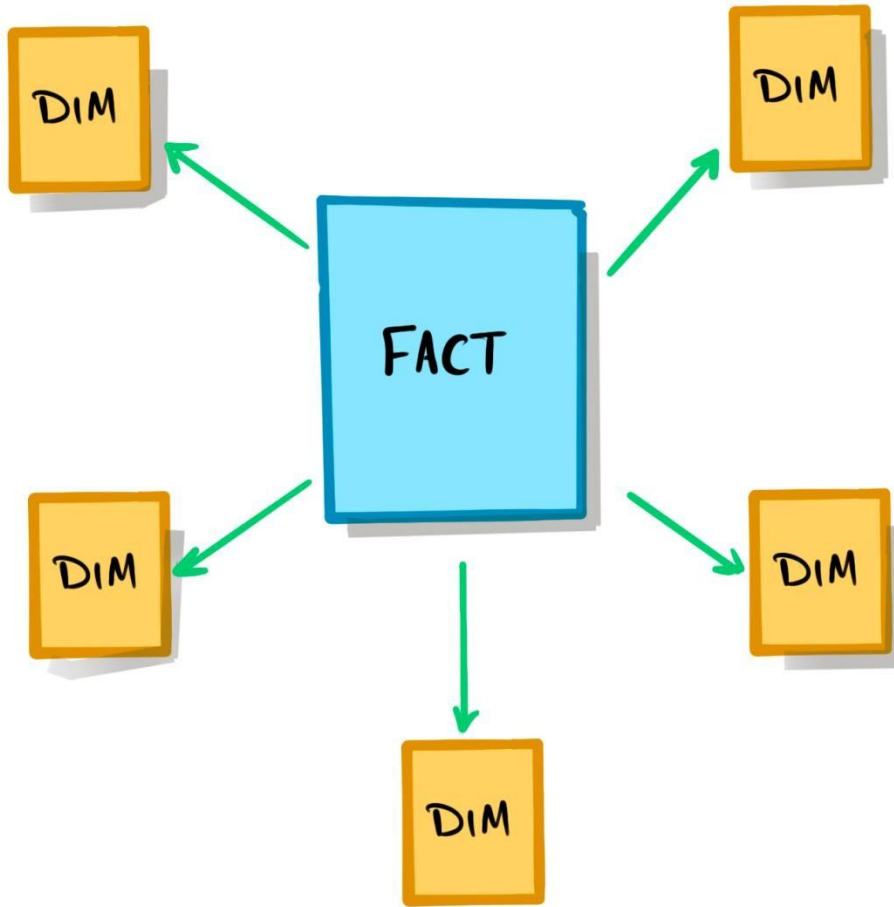




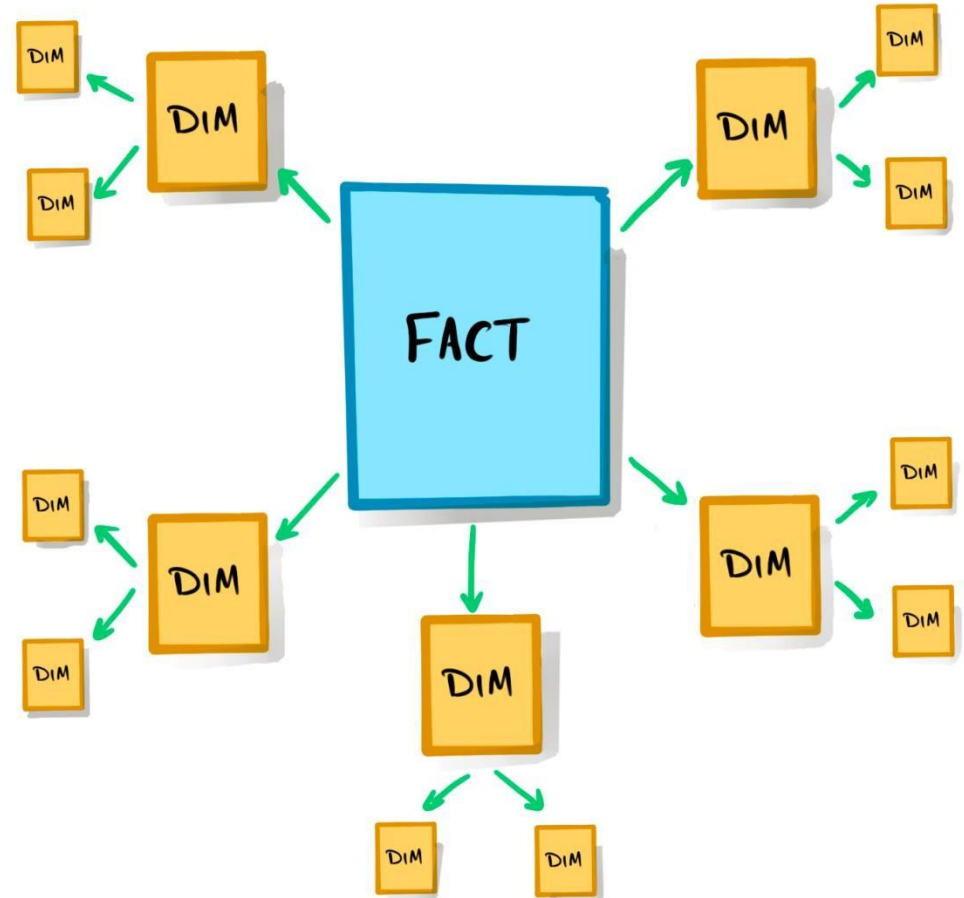
Data Architecture

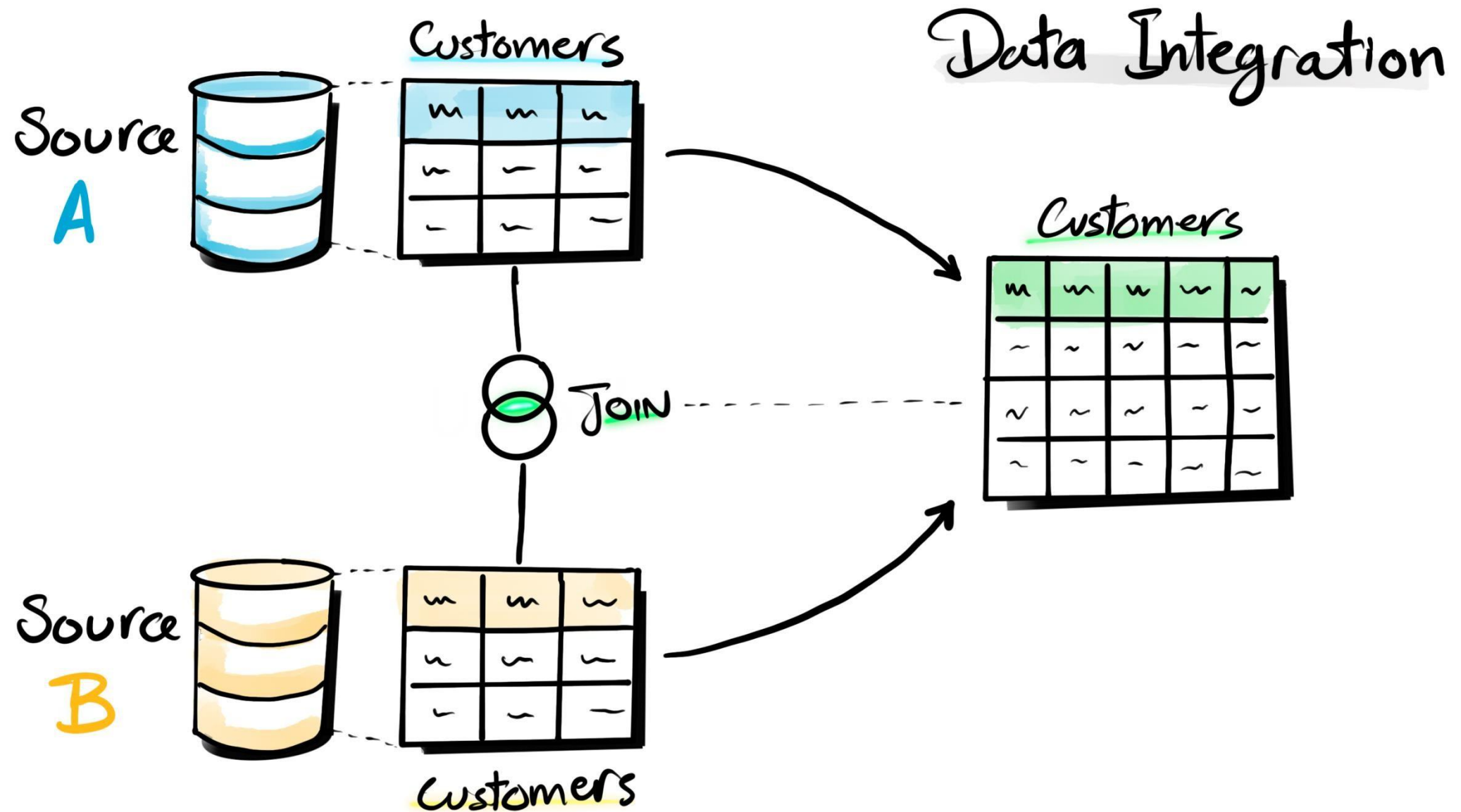


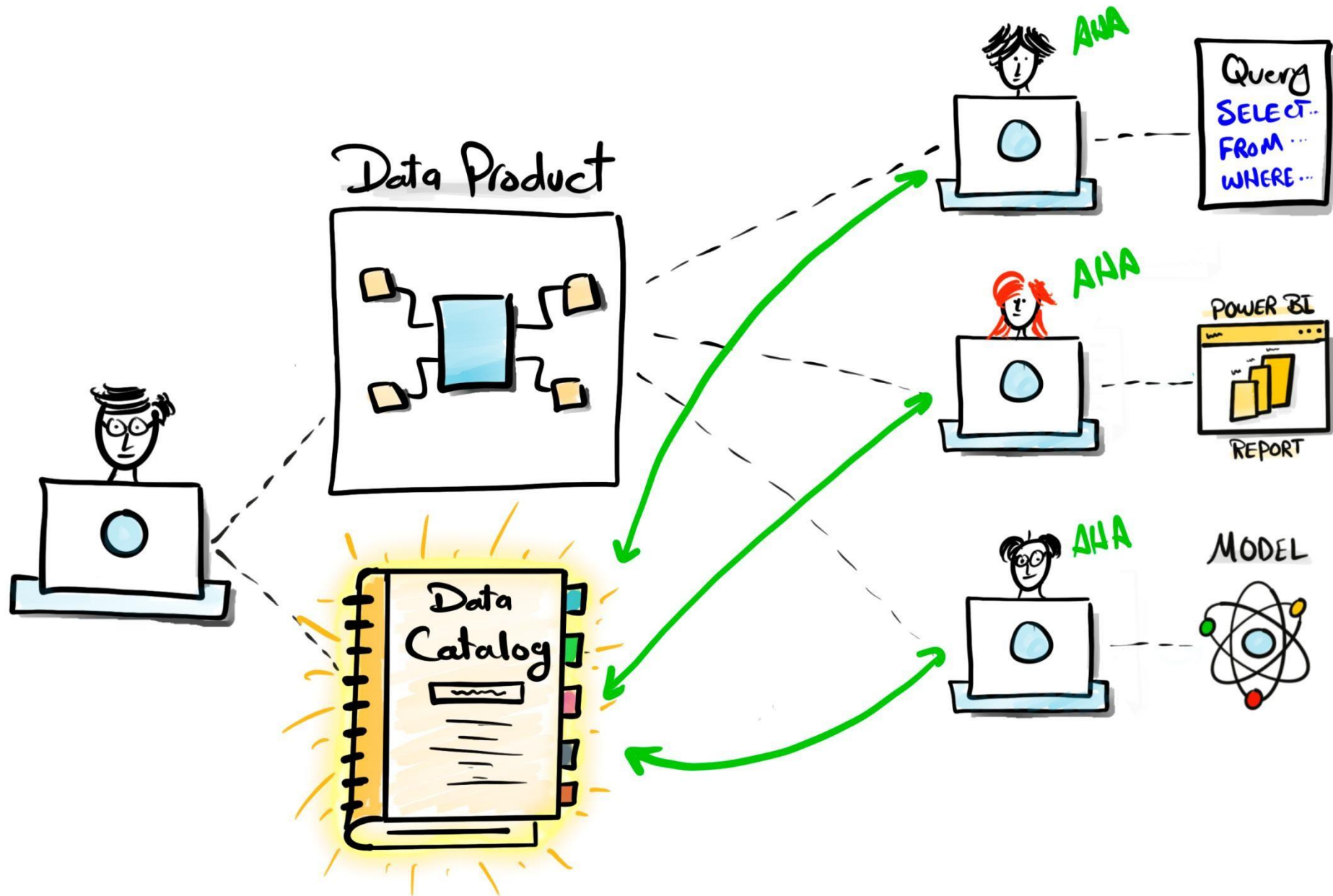
STAR SCHEMA

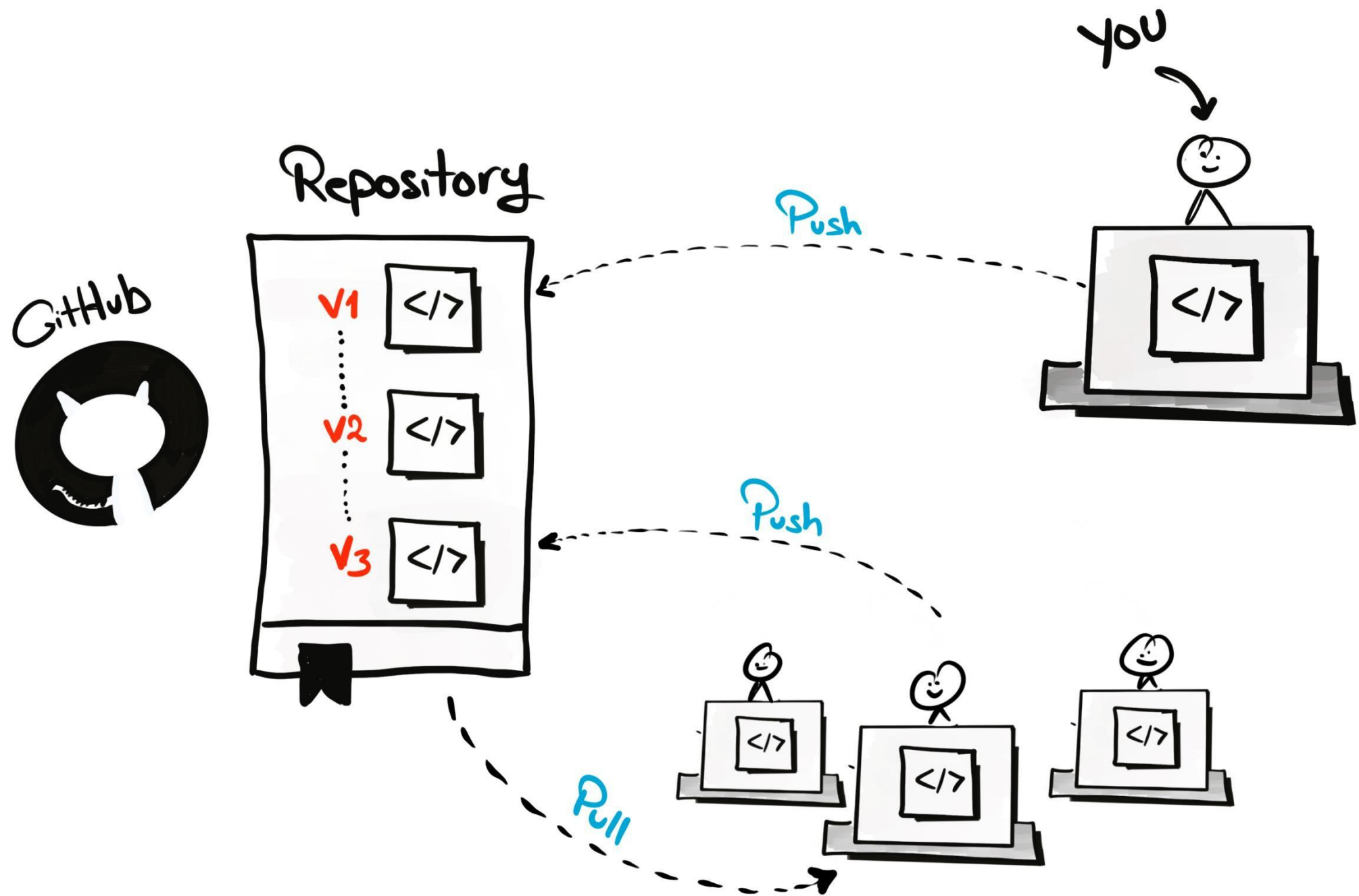


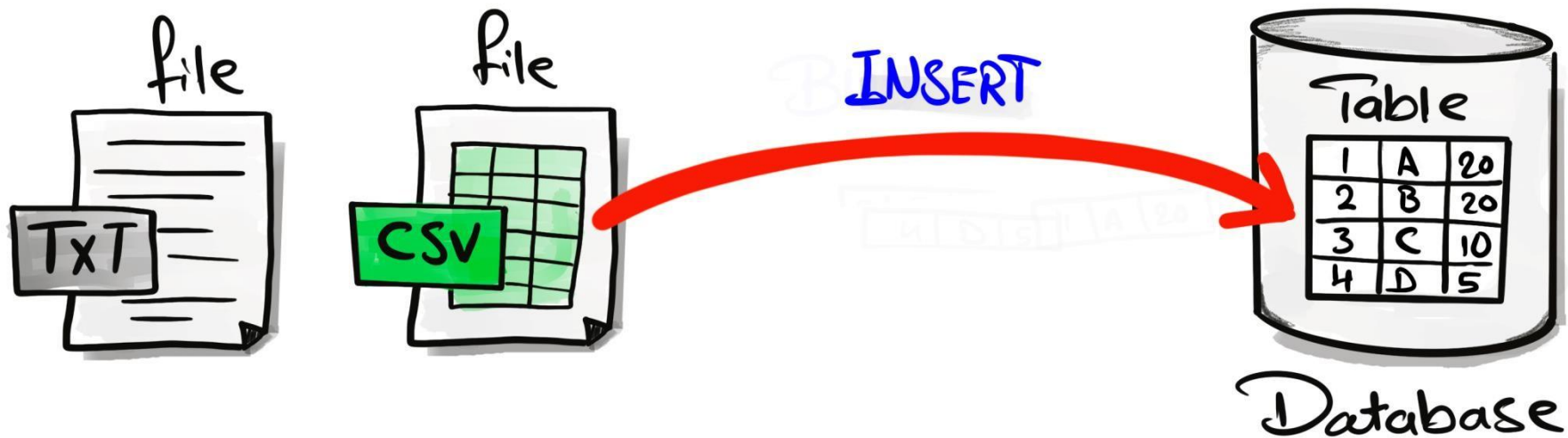
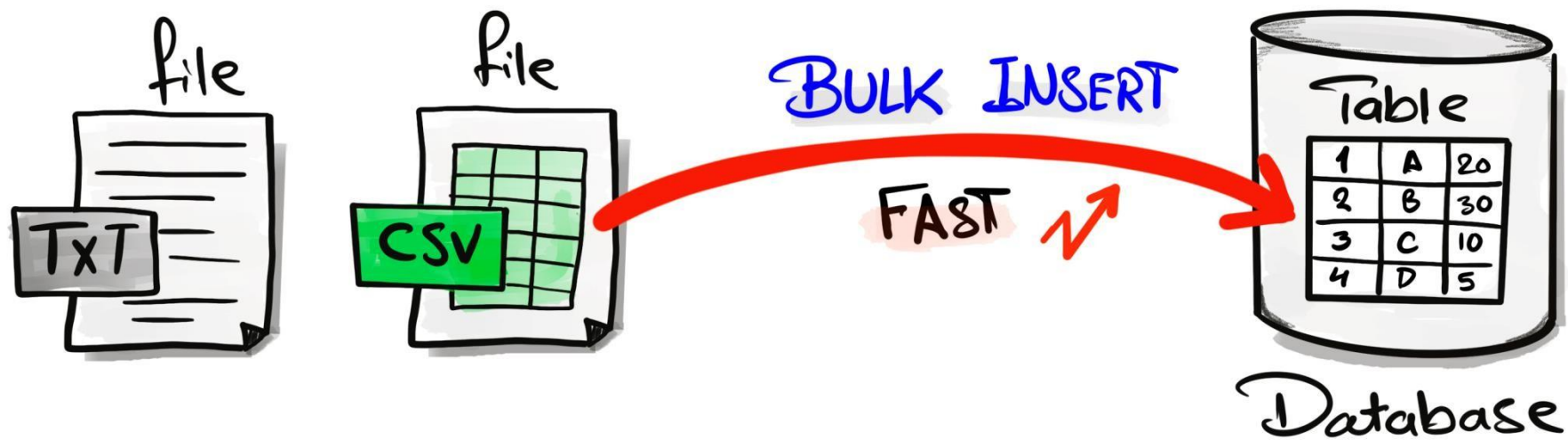
SNOWFLAKE SCHEMA













SQL DATA Analytics

Project

SQL Data Analytics Project



Dataset

~	~	~	~	~
~	~	~	~	~
~	~	~	~	~
~	~	~	~	~

Is it Numeric ?

& Does it make Sense to aggregate?

No

Yes



Dimension

Category

Products

Birthdate

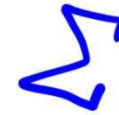
ID

Measure

Sales

Quantity

Age



A	C
B	D

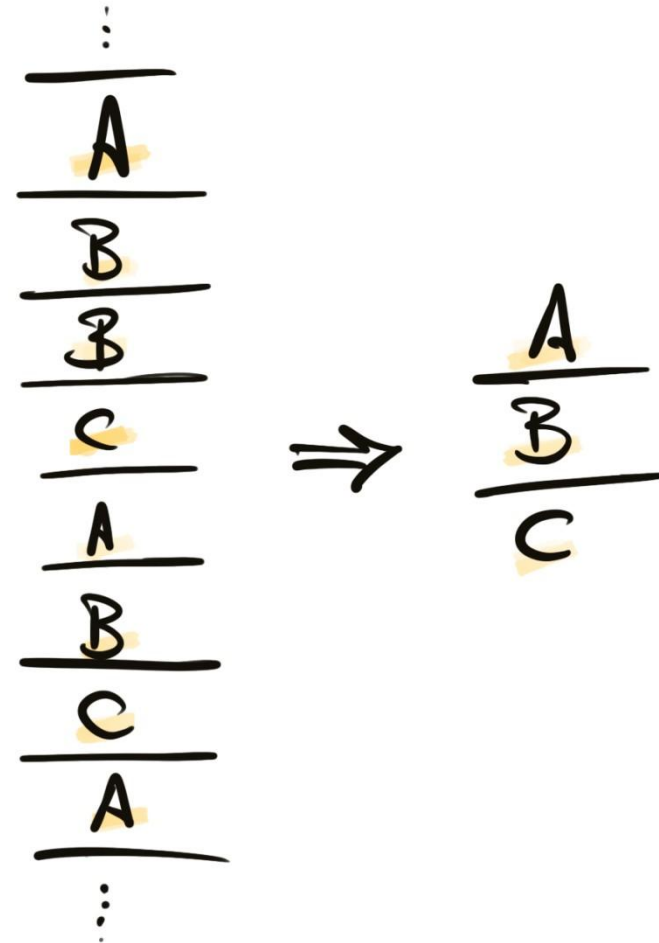
Dimensions Exploration

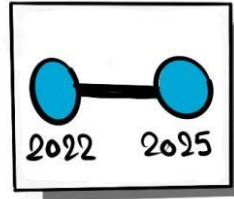
DISTINCT [Dimension]

DISTINCT Country

DISTINCT Category

DISTINCT Product





Date Exploration

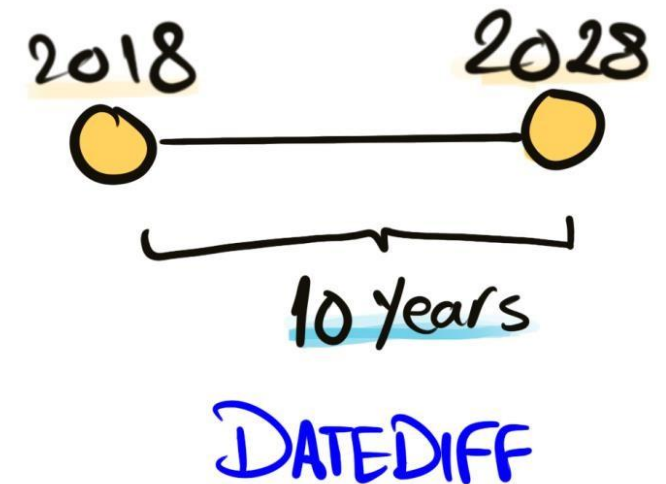
MIN/MAX [Date Dimension]

MIN Order_date

MAX Create_date

MIN Birthdate

2019
2020
2018
2018
2022
2023
2023
2028
2022



999

Measures Exploration

Σ [Measure]

SUM (Sales)

AVG (Price)

SUM (Quantity)

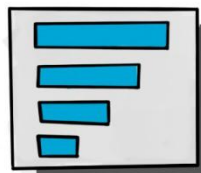
10
20
50
30
10
80
30
10

$\Rightarrow$

240

BIG Number

Key Metric



Magnitude

Σ [Measure] By [Dimension]

Total Sales By Country

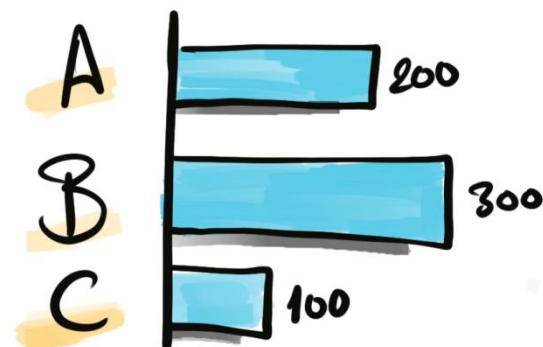
Total Quantity By Category

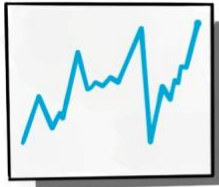
Average Price By Product

Total Orders By Customer

600

A	200
B	300
C	100





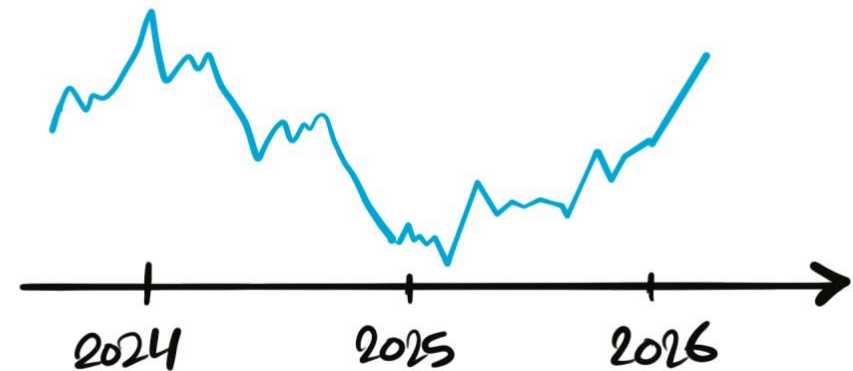
Change - Over - Time ~Trends~

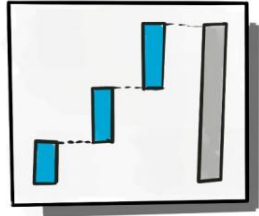
Σ [Measure] By [Date Dimension]

Total Sales By Year

Average Cost By Month

2024	300
2025	100
2026	200





Cumulative Analysis

Σ [Cumulative Measure] By [Date Dimension]

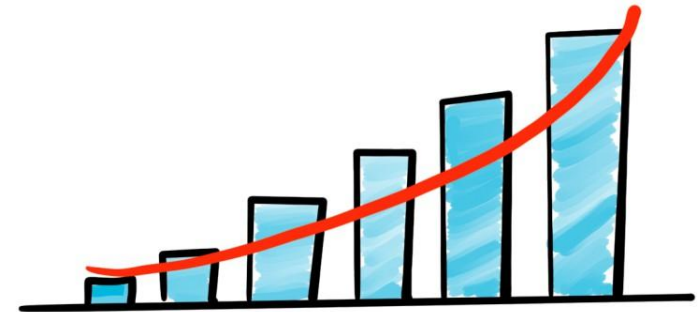
Running Total Sales By Year

Moving Average of Sales By Month

Cumulative
↓

2024	300	300
2025	100 ⁺	400
2026	200 ⁺	600

WINDOW FUNCTIONS





Performance Analysis

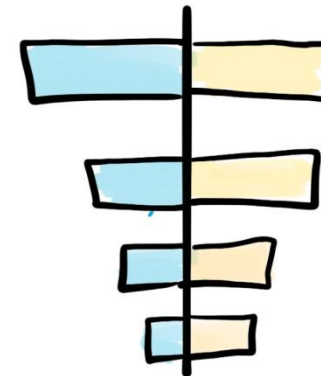
Current [Measure] - Target [Measure]

Current Sales - Average Sales

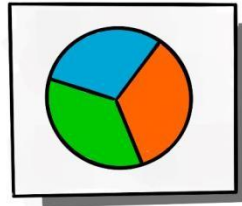
Current Year Sales - Previous Year Sales

Current Sales - lowest Sales

	Current	-	Target (AVG)	Performance
A	200	-	200	0
B	300	-	200	100
C	100	-	200	-100



WINDOW FUNCTIONS



Part-to-Whole

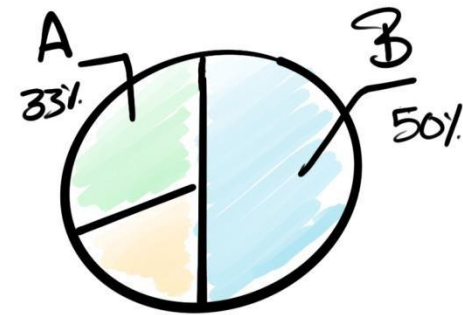
Proportional
Analysis

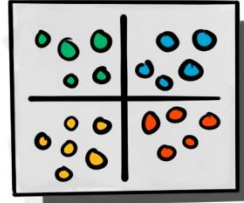
$$([Measure] / Total [Measure]) * 100 \text{ By } [Dimension]$$

$$(Sales / Total Sales) * 100 \text{ By Category}$$

$$(Quantity / Total Quantity) * 100 \text{ By Country}$$

A	200	33%
B	300	50%
C	100	17%





Data Segmentation

[Measure] By [Measure]

Total Products By Sales Range

Total Customers By Age

Σ ↘ Categorize ↙

3	50		
4	100	↘	Low
5	150	↘	Medium
1	200	↘	Large
10	250		
5	300		
			7
			6
			15

CASE WHEN STATEMENT

